Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Vortex

Selected Domain

Domain 2 - Orbyte Smart SME Challenge

Selected SDG Track (1–17)

SDG 8 - Decent Work and Economic Growth

Project / Solution Name

Vortex OrderGuard

Name (Leader)

Aun Yi Qi

Email & Phone (Leader)

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University Affiliation(s)

MMU

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Vortex is investigating whether micro and small food-service operators spend avoidable time reconciling purchase orders, delivery/receiving records and supplier invoices. Quantity, unit or price differences can require a person to locate the relevant evidence and resolve the exception before taking action. The project has not yet established how often this occurs, how much time it takes, or whether intended users would adopt a dedicated tool.

Malaysia's MSMEs contributed 39.7% of GDP and 48.7%of employment in 2025, providing economic context for studying SME productivity; these national figures do not establish demand for OrderGuard. A 2022 World Bank assessment described limits in SMEs' use of complex productivity tools and end-to-end digital transactions.

Three-way invoice matching is an established accounting control, so the proposed contribution is a focused, explainable exception-review workflow, not a new matching method.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Vortex OrderGuard is a web prototype for reviewing one purchasing item across a purchase order, receiving record and supplier invoice. It applies deterministic rules to surface quantity, invoice-quantity, price and unit exceptions; ambiguous inputs can block approval until a person clarifies them. A human confirms extracted fields and retains control of approval or rejection. Approved actions enter a simulated outbox and an activity log; the prototype does not send real supplier messages or payments. Optional AI-assisted extraction can suggest structured fields from supplier text, while a RunPod Serverless worker with EasyOCR supports OCR for synthetic invoice evidence. Where configured, the application can also call DeepSeek for text extraction. These components suggest fields only: program rules compare records and a person checks the suggestions. Manual entry remains available.The implementation uses a Flask web interface, SQLite demonstration storage, Replit deployment, and a separate RunPod GPU OCR worker. Workshop skills intended for use include web/API integration, database-backed workflows and cloud deployment. All demo business records are synthetic.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Stack: Python; Flask 3.1.3; SQLite; Bootstrap 5.3.3; Requests 2.34.2; Gunicorn 26.2.0. Replit hosts the web prototype. A RunPod Serverless GPU worker runs EasyOCR 1.7.2 for OCR; DeepSeek Chat Completions is an optional external text-extraction API when configured. Demo records are synthetic.

Flow: Synthetic PO / receiving / invoice evidence → Flask UI on Replit → optional RunPod EasyOCR and/or DeepSeek extraction → human confirms fields → deterministic matching + Trust Gate → SQLite → owner review/approval → simulated outbox + audit log. No live Orbyte API, real supplier messaging or payment processing.

SDC sponsor resource use: Team confirms SDC-provided credits/resources were used for Replit and RunPod. DeepSeek is a separate API provider, not an SDC sponsor resource

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Current Progress and Proposed Phase 2 Activities
To date, no SME interviews, design-thinking sessions, or user testing have been conducted. The team has developed four synthetic demonstration scenarios and documented a proposed testing plan. Automated regression tests have been used to verify software functionality; however, these are technical validation activities and do not constitute user research.

Proposed Phase 2 Activities (Planned)
The team plans to interview 3–5 consenting SME operators or representative users about a recent delivery-checking task. This will be followed by 12 synthetic single-item testing tasks: four clean matches, four quantity or price exceptions, and four cases involving ambiguous units.

Participants will be asked to describe their existing workflow, identify pain points, and evaluate a low-fidelity exception-review flow before testing the prototype. Data collection will include task completion, requests for assistance, corrections, confusion, and human review time.

All testing will use synthetic records to protect sensitive business information. Given the small sample size, findings will be treated as exploratory insights rather than representative conclusions about the wider SME population.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	The project aligns with SDG Target 8.2 by promoting productivity through technological innovation and improved operational efficiency, and SDG Target 8.3 by supporting productive activities and MSME growth. These represent intended contributions rather than demonstrated or achieved impacts.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	No baseline measurements have been collected to date. During Phase 2, participants will be timed while manually checking the same predefined synthetic cases, from opening the supporting evidence to reaching a final decision, including any corrections and issue resolution. The results will be reported using the median time per case, the number of tasks completed, and the sample size. No figures will be presented as measured or estimated results until supporting data has been collected.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: percentage change in median seconds per case versus manual checking, including human review and corrections. Formula: (manual median − prototype median) / manual median × 100. Also report task completion, missed exceptions and false alerts with denominators.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 Target: 20% lower median review time on the same 12 synthetic tasks, with human review included and no increase in missed exceptions. Note: This is a forward-looking target, not a result; confirm it after measuring the manual baseline and before app testing.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Flask 3.1.3; Requests 2.34.2	Web / HTTP libraries	BSD-3-Clause; Apache-2.0
Gunicorn 26.2.0; Bootstrap 5.3.3	Server / UI	MIT; MIT
RunPod SDK 1.12.0; EasyOCR 1.7.2	OCR / GPU runtime	MIT; Apache-2.0
OpenCV >=4.10, <5; Pillow >=10, <12; NumPy >=1.26, <3	Image / Runtime libraries	BSD-3-Clause; HPND; Apache-2.0

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

25/10/2026